What Does One AUC Point Buy?
Localization Ability in Weakly-Supervised Video Anomaly Detection

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Abstract

Frame-level AUC and AP are the currency of weakly-supervised video anomaly detection: leaderboards decide which model "localizes anomalies better". We show that this currency cannot buy what it is spent on. Written as a probability over frame pairs, the reported AUC decomposes exactly into three comparison sets — pairs across videos, pairs inside one anomalous video, and pairs spanning two anomalous videos — with weights fixed by the test split alone. On UCF-Crime fewer than 0.4% of pairs compare frames of the same video (0.07% on XD-Violence), so **one point of within-video localization buys 0.0039 points of reported AUC — equivalently, one reported point costs 258 (UCF-Crime) or 1,408 (XD-Violence) points of localization**. Verified on sixteen checkpoints, three consequences follow: a scorer constant inside each video attains 93.9% of a published state-of-the-art AUC while localizing at chance; the entire attainable localization range moves the reported number by 0.0019, four tenths of the seed noise; and the multiple-instance objective leaves within-video ordering a free parameter, so weakly-supervised post-processing cannot be certified to improve it. We release an estimator suite and a three-column protocol (θ, B, $B|z$) with video-clustered intervals, and report a criterion-validity check against interest: detection mAP tracks the pooled AUC ($\rho_s = 0.71$), not the localization column (0.15). We therefore offer B as a diagnostic, not a replacement headline number, and flag its external validation as open.

Keywords: weakly-supervised video anomaly detection, evaluation metrics, frame-level AUC, partial identification, sensitivity analysis

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1. Introduction

Weakly-supervised video anomaly detection trains on videos labelled only *normal* or *anomalous* and is evaluated by how well the produced frame scores separate anomalous frames from normal ones. The community reports a single number — frame-level micro-AUC (or AP) pooled over the whole test split — and publishes gains in the third decimal place as evidence of progress (UCF-Crime has moved from 75.4% to 89.4% over five years; XD-Violence AP from 73.6% to 86.9%).

Two recent audits have begun to question this practice. Song and Lee [1] observe that within-video frame pairs are a vanishing fraction of all pairs and that a video-level classifier — which produces a constant score inside each video — attains 81–97% of the reported micro-AUC while its within-video AUC is exactly 50%. Abdulaziz and Bondarev [2] show that the pooled statistic does not predict frame-level localization and that the resolution of the benchmark is too coarse for the claimed margins. Both are *empirical diagnoses*: they measure that something is wrong.

This paper does something different: it treats the reported statistic as an **estimand**, derives what it is an estimate *of*, and then asks what can and cannot be learned about localization from it. Our contributions are four.

(1) An exact decomposition of the reported metric, with an exchange rate. Pooled AUC is a probability over ordered frame pairs, and partitioning those pairs by whether the two frames come from the same video writes it as the weighted average $\theta = \omega_A A + \omega_B B + \omega_C C$ of three block statistics, with weights that are pair-count ratios — properties of the test split, identical for every method ever evaluated on it ($\omega_B = 0.0039$ on UCF-Crime, 0.00071 on XD-Violence). The identity is exact — machine precision on all sixteen checkpoints we examined — and differentiating it gives the exchange rate: **one point of within-video localization buys 0.0039 points of reported AUC on UCF-Crime, so one reported point costs 258 (UCF-Crime) or 1,408 (XD-Violence) points of localization**. Since the full attainable localization range is worth 0.0019 of θ while the seed-to-seed noise is 0.0049, **the reported metric cannot resolve localization ability at the precision this literature reports** (Section 4).

(2) Degeneracy is the optimum of the estimand, and weak labels leave localization unidentified. A scorer constant within each video attains $B \equiv 0.5$ and yet 93.9% of the published AUC of a state-of-the-art detector; the identity explains why, since such a scorer still wins all of A and most of

C, so the leaderboard's ordering is dominated by scene-level separability — a *different* capability from localization. On top of this, a multiple-instance learning (MIL) objective enters the loss only through a symmetric function of each anomalous video's frame scores, so the within-video ordering — hence B — is a free parameter of the training problem. This is a statement about the objective, not about optimization, and it predicts, correctly, that every post-processing method that uses only the weak labels fails to move B — confirmed on thirteen companion variants, a targeted counter-example, and our own attempts (Section 4.4).

(3) A causal reading of the within-video contrast, with a partial-identification fallback. The comparisons that define B hold the video fixed — scene, camera, illumination are the video's own control — so the estimand is a within-unit contrast whose identification requires that, inside an anomalous video, frame status be ignorable given observed covariates. The obvious covariate is motion energy, and adjustment alone cannot resolve whether it is confounder or mediator, since both hypotheses predict a lower estimate; stratifying, orthogonalizing, and residualizing all *decrease* B while a shuffled-motion placebo does not — a pattern consistent with mediation, on which basis the honest estimand is the unadjusted contrast (Section 5). Because ignorability cannot be verified without re-annotation, we also replace point identification by a bound: under an unknown nuisance share δ of the pair mass, the reported B certifies above-chance localization up to $\delta^* = 2B - 1$.

(4) Estimators, a reporting protocol, and a criterion-validity check reported against interest. We give the pair-weighted estimator of B, its motion-stratified variant $B|z$, video-clustered bootstrap intervals (the unit of replication is the video, not the pair), and an E-value for unmeasured confounding; a released script computes the full report $(\theta, B, B|z, \mathrm{CI}, E)$ from a score file in seconds. Re-ranking the eleven UCF-Crime checkpoints by $B|z$ moves five of them by three or more places, though the family's own spread is not resolvable as a ranking (Section 6.6). Finally, detection mAP — recomputed for twelve checkpoints under two independent implementations — tracks θ ($\rho_s = 0.71$) and not B ($\rho_s = 0.15$): the most obvious validation of our own proposal fails, and the paper says so. We therefore offer B as a diagnostic, not a replacement headline number, with its external validation flagged as open (Section 6.7). Figure 1 summarises the three facts that drive everything below: how the scored frame pairs distribute over the three blocks, how little a gain in localization is worth in the reported number, and what happens to a leaderboard when the stratified column is taken seriously.

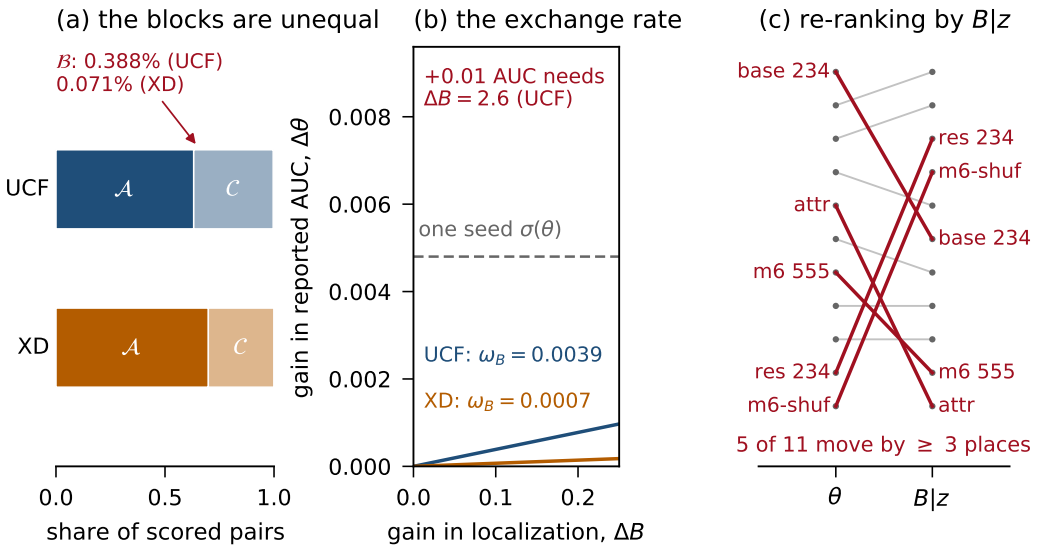


Figure 1: What one reported AUC point is made of. (a) The scored frame pairs fall into three blocks: $\mathcal{A}$ (a frame from an anomalous video against a frame from a normal video; 63.2% of pairs on UCF-Crime, 69.9% on XD-Violence), $\mathcal{B}$ (both frames from the same anomalous video; 0.388% and 0.071%), and $\mathcal{C}$ (two different anomalous videos). Only $\mathcal{B}$ can measure localization. (b) Because the reported statistic is a weighted average of the block statistics, a gain ΔB in localization is worth only $\omega_B \, \Delta B$ of reported AUC: +0.01 AUC would require $\Delta B = 2.6$ on UCF-Crime, and the seed-to-seed noise of the reported metric (0.0049) is matched only at $\Delta B = 1.26$. (c) Sorting the eleven UCF-Crime checkpoints by the stratified column $B|z$ instead of θ moves five of them by at least three places: res s234 climbs from 10 to 3, and the AUC winner base s234 falls from 1 to 6.

The rest of the paper is organized as follows. Section 2 reviews related work and Section 3 fixes notation. Section 4 states the results: the decomposition, the exchange rate, the degeneracy, the resolution bound, and the non-identifiability theorem. Section 5 turns the within-video contrast into an estimand — causal identification, the confounder-versus-mediator test, the partial-identification bound — and gives the estimators with their inference and the reporting protocol. Section 6 reports experiments on UCF-Crime and XD-Violence, Section 7 discusses what the result implies and where it stops, and Section 8 concludes.

2. Related work

Weakly-supervised VAD. The deep era began with learned temporal regularity [3]; MIL over snippet scores with a sparsity term [4] and its top-k variant [5] remain the backbone of the field. Successive refinements sharpen the objective or the features: multiple synthetic outliers as self-training supervision [6], self-supervised sparse representation [7], magnitude-contrastive glance-and-focus learning [8], uncertainty-regulated dual memory units [9], and a BatchNorm-based weakly supervised detector [10]. VadCLIP [11] replaced the feature extractor with frozen CLIP and added a language branch; DSANet [12] adds disentangled semantic alignment and is our host. All of these report frame-level micro-AUC or AP as the headline number; none decomposes it.

Metric critique. The point-adjust critique in time-series anomaly detection [13] established that a single convention can inflate scores by a large factor. For VAD, the two 2026 audits discussed in Section 7.4 document the consequences of pooling. Our contribution is orthogonal in kind: not another audit but a decomposition with weights, an exchange rate, a degeneracy construction, an identification analysis, and estimators.

Causal VAD. Lin et al. [14] cast *unsupervised* VAD as a causal inference problem with pseudo-labels as treatment and matching plus a counterfactual ensemble. Later work uses causal vocabulary for regularisation: event-relevant factor disentanglement with counterfactuals [15], causal representation consistency [16], scene-debiasing with causal normality learning [17], prototype interventions [18], and causal feature-guided augmentation [19]. We do not propose a causal *model*; we use the causal toolkit for what it is best at here — defining the estimand that the metric is a weighted average of, and separating a mediator from a confounder with a placebo intervention.

Weak supervision and identifiability. That MIL objectives identify only symmetric functions of the instance scores is folklore in the MIL literature [20], as are the noisy-OR/max identifiability discussions, and is the formal core of the UMIL analysis of instance-level bias [21]. Theorem 4 turns it into a statement about the *metric*: what the objective does not identify, the report cannot measure.

Sensitivity analysis. The E-value [22] and Rosenbaum-style bounds [23] are standard in epidemiology and rare in computer vision. We use the E-value because the contrast here is a pair-level risk ratio by construction, and because a single number ("an unmeasured within-video factor would need a 4.5-fold association with both status and score") is more useful to a reader

than an unqualified claim of robustness.

Decomposition of AUC. Splitting a pooled AUC into within-group and between-group components is classical in the discrimination literature (the within/between variance decomposition behind multiclass AUC, [24]). What is new here is not the algebra but its use as a *measurement instrument*: computing the weights from the split, deriving the exchange rate from them, bounding what any method can achieve, and defining an estimator for the component the field believes it is reporting.

3. Setup and notation

Notation. We use one letter for one thing. s is the scorer, $s_i = s(x_i)$ the score of unit i, and S_i the corresponding random variable; $y_i \in \{0, 1\}$ is the ground-truth label of unit i and y_v the label vector of video v; a_v and b_v count the anomalous and the background units of an anomalous video v. θ is the reported statistic of Eq. (1); A, B, C are the block statistics of Theorem 1 and $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ the corresponding pair sets. B is also our name for the localization estimand (they coincide by construction), $B|z$ is its motion-stratified version (Section 5.6), $D = B - B|z$, and a prefix Δ always means the difference between two scorers on one split ($\Delta\theta = \theta - \theta'$ and so on). Intervals are 95% video-clustered bootstrap confidence intervals (CI), E is the E-value of Section 5.8, ρ_s is Spearman's rank correlation across a family of checkpoints together with its two-sided p-value, and δ is the share of contaminated pair mass of Section 5.3; $\pi_{\text{mot}} = (B - B|z)/(1 - B|z)$ is the motion-mediated share of the within-video signal (a mediation proportion, Section 5.3).

Data. A test split is a collection of V videos; video v contributes n_v scored units (snippets), each unit carrying a binary label $y_i \in \{0, 1\}$, and $y_v \in \{0, 1\}^{n_v}$ collects the labels of one video. Let $\mathcal{P}$ be the set of *ordered* frame pairs (i, j) with $y_i = 1$, $y_j = 0$, and $N = |\mathcal{P}|$. Let $\mathcal{A} \subseteq \mathcal{P}$ be the pairs whose negative member lies in a *normal* video, $\mathcal{B} \subseteq \mathcal{P}$ the pairs whose negative member lies in the *same* anomalous video as the positive member, and $\mathcal{C} = \mathcal{P} \setminus (\mathcal{A} \cup \mathcal{B})$ the pairs whose negative member lies in a *different* anomalous video. For a scoring function s, the reported statistic is

$$\theta = \frac{1}{N} \sum_{(i,j)\in\mathcal{P}} \Big(\mathbb{1}[s_i > s_j] + \tfrac{1}{2}\mathbb{1}[s_i = s_j] \Big), \tag{1}$$

the Mann–Whitney form of frame-level micro-AUC [25] (ties contribute one half, equivalently the average-rank convention; AP is handled in Section 4.5).

Table 1: Pair census of the two benchmarks, computed from the ground-truth arrays alone. The weights are properties of the label layout, not of any model.

split	videos	frames	positives	ω_A	ω_B	ω_C	n_B/N	$1/\omega_B$
UCF-Crime test	290	69,368	5,260	0.6317	0.00388	0.3645	0.388%	257.7
XD-Violence test	800	2,331,296	537,882	0.6985	0.00071	0.3008	0.071%	1,407.6

Write $n_A = |\mathcal{A}|$, $n_B = |\mathcal{B}|$, $n_C = |\mathcal{C}|$, and let A, B, C be the same statistic restricted to each block. Then $\omega_Z = n_Z/N$ for $Z \in \{A, B, C\}$.

Why these three blocks. $\mathcal{A}$ compares an anomalous frame with a frame from a video that is entirely normal: it can be won by recognizing the *scene* (a highway vs. a mall), since normal videos and anomalous videos are different recordings of different places. $\mathcal{B}$ is the only block in which both frames come from the same recording, so scene, camera, illumination, and encoding are held fixed by construction: it is the only block that can measure *localization*. $\mathcal{C}$ again crosses videos, but both sides are anomalous recordings, so it measures relative severity rather than normality.

The pair census is data-only. ω depends on the label layout of the split, not on any model. Table 1 lists the census for the two standard benchmarks [4, 26].

Two facts are worth stating before any theory. First, ω_B is about two orders of magnitude smaller than the other two weights. Second, ω_B shrinks as the benchmark grows ($0.0039 \to 0.0007$) because the number of same-video pairs grows only linearly in the video length, while the cross-video pairs grow quadratically: **the larger the benchmark, the less the reported number is about localization.** Song and Lee [1] report the same fraction in the range 0.071–0.388% across five benchmarks; our census reproduces both endpoints of that range, UCF-Crime at 0.388% and XD-Violence at 0.071%.

The estimand. We define the *localization ability* of a scorer as $\tau(s) = B$, the probability that, inside an anomalous video, a randomly chosen anomalous frame is scored above a randomly chosen normal frame of that same video — a within-unit contrast whose control group is supplied by the video itself. If a scorer is to be credited with localization, τ is the quantity that must move; θ is not.

4. What the reported number is a measure of

Throughout, s is any real-valued scorer of the scored units, ties contribute one half as in Eq. (1), and ω_Z are the split-level pair weights of Section 3.

4.1. Theorem 1 (exact decomposition)

For every scorer s,

$$\theta = \omega_A A + \omega_B B + \omega_C C, \qquad \omega_A + \omega_B + \omega_C = 1. \tag{2}$$

Proof. $\mathcal{P} = \mathcal{A} \sqcup \mathcal{B} \sqcup \mathcal{C}$ is a partition, and θ is an average over $\mathcal{P}$. Grouping the sum by block gives the result, with A, B, C the block averages. The weights are pair-count ratios and do not involve s. ■

The identity is not an approximation: we verified it to 0.00000 on eleven UCF-Crime and four XD-Violence checkpoints (maximum absolute deviation $< 10^{-12}$) and on every probe below. It is also *not* the same statement as "AUC is dominated by cross-video pairs" — it says exactly how much, and it separates the two different cross-video mechanisms (A: scene/normality; C: relative severity). The decomposition is an accounting identity, not a causal attribution; the danger of reading an aggregate association as a component effect is the classical one [27].

Corollary 1 (exchange rate). $\partial\theta/\partial B = \omega_B$. On UCF-Crime one point of B buys 0.0039 points of θ; equivalently **one reported point costs 257.7 points of localization** on UCF-Crime and 1,407.6 on XD-Violence. Since $(\omega_A + \omega_C)/\omega_B = 256.7$ (UCF) and $1,407.6$ (XD), a method that raises *scene-level* discrimination by one point gains what a method that raises localization by 257 (UCF) or 1,408 (XD) points gains.

4.2. Theorem 2 (degeneracy: the constant-within-video scorer)

Let s be constant inside every video, $s_i = f(v)$ for all $i \in v$. Then $B(s) = 1/2$ exactly, and

$$\theta = \omega_A A_f + \omega_C C_f + \omega_B/2, \tag{3}$$

with A_f, C_f the block statistics of the video-level score f. Consequently, *among scorers that are constant inside each video*, $\sup \theta = \omega_A + \omega_C\, C^* + \omega_B/2$, where C^* is the largest pair-weighted concordance that any ranking of the anomalous videos attains on $\mathcal{C}$. $C^* < 1$ strictly: for every ordered pair of distinct anomalous videos, both directions lie in $\mathcal{C}$, and at most one of them

can win. (Over all scorers the supremum is of course 1, attained by a perfect localizer.)

Proof. For any pair inside the same video the two scores are equal, so the indicator is a fair coin: $B = 1/2$. The second display is Eq. (2). For the supremum: $A_f \leq 1$, attained by ranking every anomalous video above every normal one; subject to that, θ is maximized by maximizing the weighted concordance C_f over rankings of the anomalous videos, a finite linear-ordering problem whose maximum C^* is strictly below 1 by the two-direction argument above. ■

Theorem 2 is the mechanism behind the empirical observation of Song and Lee [1]: a scorer with *no* localization ability whatsoever is not far below the state of the art, because the only block it forfeits entirely is $\mathcal{B}$ — a video-level score can still win all of $\mathcal{A}$ and most of $\mathcal{C}$ (the classifier below attains $C_f \approx 0.56$). Measured: a linear video-level classifier trained on cached CLIP features reaches $\theta = 0.8234$ with $B = 0.5000$ exactly, against $\theta = 0.8257$ for a frame-level MIL model trained identically ($B = 0.657$) and 0.8772 for the published DSANet checkpoint. The video-constant scorer thus keeps **99.7% of the identically-trained frame model's θ and 93.9% of the published DSANet checkpoint's** θ — and 92.1% of the best number reported in the literature (89.4%) — while its localization term is exactly chance.

Note what Theorem 2 does *not* say: it does not say the constant scorer has a high B, nor that θ is a poor measure of *anomaly detection in general*. It says that θ is an average over three comparison sets of which one — the only one that can see localization — carries 0.388% of the weight.

4.3. Theorem 3 (the inversion is uninformative and unusable)

Suppose a paper reports θ and θ' for two scorers and claims the difference is due to localization. By Theorem 1,

$$\Delta B = (\Delta\theta - \omega_A \Delta A - \omega_C \Delta C)/\omega_B. \tag{4}$$

Two things follow.

(a) Non-identification. Without separately measuring ΔA and ΔC, a difference $\Delta\theta$ is compatible with *any* $\Delta B \in [-1, 1]$: choose ΔA accordingly ($\Delta A = (\Delta\theta - \omega_B \Delta B - \omega_C \Delta C)/\omega_A$, which is feasible for every ΔB because $\omega_A = 163\,\omega_B$). A reported AUC difference therefore carries no information about which block improved until A and C are reported.

(b) Error amplification. If $\Delta\theta$ is measured with standard error σ, the implied $\Delta\hat{B}$ has standard error $\sigma/\omega_B = 258\sigma$ (UCF) and $1{,}408\sigma$ (XD). With the seed-to-seed $\sigma(\theta) = 0.0049$ that we measure on our checkpoints (Section 6.4), a single-seed comparison gives $\mathrm{SE}(\Delta\hat{B}) = 1.26$: the 95% interval implied for ΔB is about ten times the width of B's entire range (0.5), so the reconstructed quantity is not even bounded by the parameter space. To certify a one-point localization gain ($\Delta B = 0.01$) with a two-standard-error margin one would need $\sigma(\Delta\theta) \leq 1.9 \cdot 10^{-5}$ — a standard error 253 times smaller than the observed seed noise, which at $\sigma \propto k^{-1/2}$ would take roughly $253^2 \approx 64{,}000$ seed replicates.

Proof. Algebra of Theorem 1; feasibility of ΔA uses $\omega_A > 0$; the amplification is $1/\omega_B$. ■

Theorem 3 is why we do not merely *report* B alongside θ: we argue that B cannot be *recovered* from reported θ differences, so the community's practice of reading localization progress off the AUC column is not merely noisy but structurally empty. The only remedy is to measure the blocks.

4.4. Theorem 4 (weak labels do not identify the within-video ordering)

Let the training loss $\mathcal{L}$ depend on the frame scores of anomalous video v only through a symmetric function of the multiset $\{s_i : i \in v\}$ — as every MIL objective does (top-k mean, maximum, attention pool, expectation over a categorical head). For any video-wise permutation π that keeps the labels of the frames fixed, $\mathcal{L}(s \circ \pi) = \mathcal{L}(s)$. Hence the minimizer set of the training problem is closed under such permutations, and B is not a functional of the labels: **for the same minimal loss, every ordering of the frames inside a video is admissible**, including ones with $B = 1$ and ones with $B = 0$. The scope of the statement is set by its hypothesis: it covers objectives all of whose terms are functions of the multiset of scores, and it excludes any term that couples frames at different time indices — in particular the temporal-smoothness penalty $\sum_t \|s_{t+1} - s_t\|^2$ used by Sultani et al., which is *not* permutation-invariant and therefore does constrain the ordering.

Two things delimit the claim, and we prefer to state both here than to have a reader find them.

Objectives with time coupling are outside the theorem. A temporal-smoothness penalty on the scores, or any other term that depends on where a frame sits in the sequence, breaks the permutation invariance and pins part of the ordering. The original MIL objective of Sultani et al. contains such a term, so the theorem does not cover it; what covers it is the empirical record

of item (ii) below, which tests objectives with and without such regularisers and finds the ordering unmoved either way.

The premise concerns the value of the loss, not its gradient. The top-k-minus-mean contrast T_v of remark (ii) is symmetric *as a functional* — it is a function of the multiset of scores — so the theorem does apply to it: every permutation of the frames attains the same loss and the minimizer set remains permutation-closed. What is asymmetric is where the gradient flows, and that does select an ordering during optimization. We count that, and every regulariser with the same property, as part of a method's inductive bias rather than as identification: nothing in the weak labels makes the ordering it selects correct, and the constraint is itself an assumption about anomalies (that they are a sparse, top-scoring sub-segment) that the annotation does not verify. The empirical claim — that published objectives do not pin the ordering — is unaffected.

Proof. Symmetry of the pooling function gives equality of the loss term, and the same argument applies term by term to every other term that is a function of the multiset of scores; by the hypothesis the objective contains no term of any other kind. Since B is a function of the pairwise ordering, the claim follows. ■

Three remarks.

(i) Theorem 4 is a statement about the *objective*, not about optimization: SGD plus architectural inductive bias (temporal attention, positional encodings) does impose *some* ordering, but nothing in the objective rewards the ordering being correct.

(ii) It explains a body of negative results. A companion study [28] tried thirteen post-processing schemes on top of weak-label training (halo suppression, IPW reweighting, score-invariant reordering, oracle recalibration with ground-truth distances, ...); all thirteen failed, and an oracle that reweights by the *true* distance to the annotated segment was 0.0045 AP *worse* than no correction. An independent attempt of ours — training an auxiliary head to make the top-k-minus-mean contrast $T_v = \mathrm{top}_k(s) - \mathrm{mean}(s)$ — where top_k is the mean of the k largest scores of a video and mean the mean of all of them — positive on anomalous videos fails for a structural reason visible in one line: $T_v \geq 0$ always, so the constraint it imposes on normal videos ($y_v = 0 \Rightarrow T_v \leq m$, for a margin m) is unsatisfiable, and the arm ends with $B = 0.3124$ and 81% of the videos ranked *backwards*.

(iii) The corollary is a design constraint, not a counsel of despair: it says that localization must be obtained from something *other* than the weak labels — architecture, auxiliary modalities, or a training signal that does not collapse the frames into a symmetric statistic. It also says that any claimed localization improvement must be *measured* on B, since the labels could not have produced it.

4.5. What about AP?

Average precision is not additive over blocks, because precision is computed along the global ranking and interacts with the pool of negatives as a whole. We therefore do not claim an exact law for AP, but the qualitative statement is stronger rather than weaker: AP is an average over the same pool of positive-negative pairs but weighted by the rank positions of positives, and the n_B/N share is the same 0.388% (UCF) / 0.071% (XD), so the arguments of Sections 4.1–4.4 about which pairs the statistic can see transfer qualitatively. The pair-weighted view of AP has a precedent in precision at a recall target [29], where the same pool-dependence is why a single scalar is not additive. What does not transfer is the exact law: AP's weights depend on rank positions and therefore on the whole negative pool, so an AP version of Theorem 1 would need a price-weighted decomposition that we do not attempt. Empirically the two statistics agree in ordering on the checkpoints we have: the four XD-Violence checkpoints of Section 6.9 have frame AP 0.8113–0.8658 and micro-AUC 0.9368–0.9536, and the two orderings coincide exactly. We keep the theory to AUC and say so wherever AP is used.

5. Identifying, bounding, and estimating the within-video contrast

Theorems 1–4 say what to measure. They do not say whether the measured B is the quantity we want, because B is an *observational* contrast: inside an anomalous video the frames labelled anomalous differ from the frames labelled background in many ways, and the scorer may be responding to any of them. This section makes the target explicit and then tests the threat that dominates the discussion in practice — motion.

5.1. The contrast as a causal estimand

Fix a *frozen* scorer s and an anomalous video v with a_v anomalous and b_v background units. For a unit i let $y_i \in \{0, 1\}$ be its (annotated) status and $S_i = s(x_i)$ its score. The within-video contrast is

$$\tau_v = P(S_i > S_j \mid y_i = 1, y_j = 0,\ i, j \in v), \tag{5}$$

and the localization term is the pair-count weighted average $B = \sum_v w_v \tau_v$, $w_v \propto a_v b_v$. Two features distinguish τ_v from the usual detection estimand. First, the video is its own control: scene, camera, illumination and encoding are held fixed by construction, which is exactly why $\mathcal{B}$ is the informative block. Second, the unit of replication is the *video*: τ_v is estimated inside one video, so uncertainty must be assessed by resampling videos (Section 5.7), not pairs. Identification of τ_v requires four assumptions:

- **A1 (no interference / well-defined unit score).** S_i depends on unit i only. This holds for a post-hoc scorer applied unit-wise, and it is what a deployed detector does *within a window*, but it does **not** hold for the training objective of a MIL model, where the top-k/max pool makes the loss depend on the joint configuration of scores (Section 4.4). We therefore estimate B for frozen checkpoints at inference time and we flag that the score of a frame produced by window attention is itself a joint quantity.
- **A2 (within-video ignorability).** Given a covariate Z that varies inside the video, the status of a unit is independent of its score's rank:

$$y_i \perp \mathrm{rank}(S_i) \mid Z_i, v. \tag{6}$$

- **A3 (positivity).** $a_v, b_v > 0$ — true by construction of $\mathcal{B}$.
- **A4 (no annotation error).** The labels equal the true status.

Three of these are known to be strained in this literature and we prefer to name them than to hide them. A1 is violated by design at training time (Section 4.4). A3 holds for UCF-Crime and XD-Violence but *not* for the 150 UCF-Crime test videos that are entirely normal (51.7% of the split): those videos contribute to $\mathcal{A}$ but cannot contribute to $\mathcal{B}$, so B is estimated on a biased subset of the split — the anomalous ones. A4 is strained by the coarse, segment-level annotation of both benchmarks: a companion study [28] found that 95–97% of the AP loss of a trained detector comes from "halo" frames — normal-looking frames inside anomalous videos that the model scores high — and that these are partly *genuinely* anomalous frames outside the annotated segment; reordering frames by their distance to the annotated segment made AP *worse* ($-$ 0.029). If A4 fails, B is biased *downward* (true

anomalies are counted as controls), so our estimates are conservative. A2 is the one assumption that no observable feature of the data can check, so we do not ask the reader to grant it: Section 5.3 replaces point identification by an interval whose width is a nuisance share the reader chooses.

5.2. Motion: confounder, mediator, or neither?

The canonical nuisance inside a video is motion energy. The argument for treating it as a confounder is direct: anomalies involve motion, and a scorer sensitive to motion energy will rate any high-motion frame highly, so part of B could be pure motion response rather than anomaly evidence. If motion is a confounder — a common cause of the label and the score — the observed contrast is inflated, and adjusting for it should move the estimate *down* toward the direct effect. The alternative is that motion is on the causal path — the anomaly *causes* motion, which *causes* the score — in which case adjusting also moves the estimate down, by removing signal. This is the confounder-versus-mediator fork of causal adjustment [30]. The sign of the adjustment response therefore cannot by itself separate the two hypotheses; what the design below tests is whether the response is specific to real motion structure, and even that test is one-sided, because the placebo of item (3) is a weak control: a nuisance permuted across videos is nearly uncorrelated with the score by construction.

Stratification alone cannot separate the two, because the adjustment is the same operation. We therefore run three interventions, one of which is a placebo:

1. **Stratify** (observational): compare anomalous and background units only within the same motion quintile inside the same video ($B|z$, Section 5.6).
2. **Intervene on training** (experimental): add to the loss a penalty on the squared within-video score–motion correlation, with weight γ: $\gamma \,\mathrm{mean}_v \mathrm{corr}^2(S_v, Z_v)$ (Section 6.5).
3. **Placebo**: the same penalty applied against *shuffled* motion — Z is permuted across videos so that the penalty acts on the wrong video's motion profile. If the effect of (2) is due to decorrelating against real anomaly-related structure, (3) must be inert; if it is a generic regularisation effect, (3) will look like (2).

The result (Section 6.5) is a consistent pattern: (1) lowers B by 3.5–5.8 points; (2) lowers it monotonically in γ; and (3) is inert — it recovers the

unregularised level (0.689 against a baseline of 0.680 for B, 0.648 against 0.643 for $B|z$). Both hypotheses predict a drop under (1) and (2), so the sign alone does not decide between them; what the pattern establishes is that the response is specific to real motion structure rather than to regularisation as such. We read motion as a **mediator** — the annotated events are largely motion-defined, so the path anomaly → motion → score is the natural one — and on that reading adjusting for it is over-adjustment: the honest target is the *unadjusted* B, and the adjusted $B|z$ is a diagnostic of how much of the localization signal is motion-mediated. The size of that adjustment is the gap $D = B - B|z$, which is 3.5–5.8 points (median 4.9) across the eleven UCF-Crime checkpoints.

The contrast with a confounder is worth stressing, because the instinct in this literature is to "de-bias" against motion: had motion been a confounder, D would be the *spurious* part and $B|z$ the estimand of interest. We flag two limits of the verdict. First, the placebo is a weak control for the reason given above, so its inertness rules out less than the design suggests. Second, the mediator reading predicts that the gap D should grow with the score–motion correlation; across the eleven checkpoints it does not (Spearman $\rho \approx -0.08$). The mediator reading remains the more natural one, but it is a modelling assumption we make and state, not a causal verdict the design delivers.

Two further diagnostics agree. A zero-training motion statistic — the 5-frame moving average of $\|\Delta x_t\|$ on frozen features — attains $B = 0.6552$ and $B|z = 0.5465$, i.e. it accounts for $(0.6552 - 0.5)/(0.6803 - 0.5) = 86\%$ of the within-video signal of a trained frame-level probe and for *nothing* after stratifying on itself. And the remaining coupling in trained models is small but non-zero: the within-video correlation between score and motion is 0.12–0.33 across the eleven UCF-Crime checkpoints (mean 0.18). A scorer is not a motion detector, but in the only block that measures localization it is close to one.

5.3. When A2 fails: a partial-identification bound

A2 is the load-bearing assumption of the whole construction, and it is not verifiable without re-annotating the benchmark: nothing in the data tells us whether the background frames of a violent video are unremarkable *apart from* the violence. Rather than assert it, we replace point identification by an interval and let the reader choose the violation. Model the violation as contamination of the pair mass: within each video, an unknown fraction δ of positive–background pairs is ranked by a video-level nuisance (temporal

position, exposure, camera drift, annotation spill-over) rather than by the anomaly, and for those pairs the comparison may go either way; the remaining $1-\delta$ are ranked as the causal contrast τ would rank them. (Bounds of exactly this kind are the standard substitute for unverifiable point identification in the partial-identification literature; see [31] for the original worst-case construction in a treatment-effect setting.)

Proposition 5 (partial identification of τ under A2 violation). Under A1, A3, A4 and δ-contamination as above, with B the reported within-video contrast,

$$\tau_{\mathrm{LB}}(\delta) = \max\left\{0, \frac{B-\delta}{1-\delta}\right\} \;\le\; \tau \;\le\; \min\left\{1, \frac{B}{1-\delta}\right\} = \tau_{\mathrm{UB}}(\delta), \qquad 0 \le \delta < 1. \tag{7}$$

Proof. Conditioning on which pairs are contaminated, $B = (1-\delta)\tau + \delta q$ with $q \in [0,1]$ the (unknown) rate at which the contaminated pairs happen to be ranked correctly; solving for τ over the range of q gives the two endpoints. Both endpoints are then intersected with the unit interval, the range of τ: the upper one because $\tau \le 1$, the lower one because a nuisance share above B leaves no information and the bound correctly degrades to the trivial $0 \le \tau$. ■

The bound is stated under A1, and A1 is the one assumption the host does not satisfy: a frame score produced by attention across the window is a function of the whole video rather than of its own frame. Interference of that kind can move a given pair either way, so the interval is not a certificate for the attention-based checkpoints we actually measure; it is the bound a unit-wise scorer would earn, and for our host it should be read as the tolerance $\delta^* > 1/2$ of a *fixed* window-level aggregation rather than as evidence about localization. Three further limitations of the reading appear in Section 7.6.

Three consequences make the bound useful rather than decorative.

Identification power, per checkpoint. $\tau_{\mathrm{LB}}(\delta) > 0.5$ — the claim "this model ranks anomalous frames above its own video's background better than chance" — holds exactly while $\delta < 2B - 1 =: \delta^*$. So every reported B carries its own tolerance: a nuisance share below δ^* cannot explain the result away, one above it can. On the eleven UCF-Crime checkpoints $\delta^* \in [0.416, 0.515]$ (median 0.447) and on the four XD-Violence checkpoints $\delta^* \in [0.609, 0.693]$, i.e. the localization claim survives unless roughly 40% (UCF) to 60% (XD) of the within-video pair mass is nuisance-driven.

What the measured nuisance does and does not buy. The only nuisance we can quantify is motion, and the stratified estimator attaches to it the mediation proportion $\pi_{\text{mot}} = (B - B|z)/(1 - B|z) \in [0.120, 0.170]$ (median 0.149) on UCF and $[0.122, 0.141]$ on XD. This is *not* a share of contaminated pair mass and must not be compared with δ^*: it is derived from the mixture model by setting $q = 1$ — contaminated pairs ranked correctly — the endpoint most favourable to the claim, so it measures how much of the *signal* is motion-carried, not how much of the pair mass a nuisance governs. A nuisance that threatens the localization claim lives at the opposite end of the mixture (q small), and no measured candidate populates that end. The bound, not the measured share, is the tool for unnamed nuisances.

Monotonicity. τ_{LB} is increasing in the reported statistic — it is affine in B with slope $1/(1 - \delta) > 1$ — so the bound cannot invert the ordering of checkpoints. What it does to differences is the opposite of compressing them: pairwise gaps are *stretched* by $1/(1 - \delta)$, while the admissible range shrinks to $[(1/2 - \delta)/(1 - \delta), 1]$, which at $\delta = 0.2$ is $[0.375, 1]$ and still places the estimates at 0.635–0.697 (UCF) and 0.756–0.808 (XD), well above chance.

Table C.8 (Appendix Appendix C) collects the numbers per checkpoint. The bound is deliberately conservative in one direction only: contamination can push B *up* as well as down, which is why a high B under an unknown δ must not be read as a high τ unless $\delta < \delta^*$.

We check the algebra and the sharpness of the bound by simulation (Appendix A): with $m = 400$ positives, $n = 4{,}000$ negatives and a known $\tau \approx 0.760$, pushing the δm contaminated positives above every negative gives observed $B = 0.785/0.809/0.832/0.880$ for $\delta = 0.1/0.2/0.3/0.5$, and the interval $[\tau_{\text{LB}}, \tau_{\text{UB}}]$ contains the truth in every run, with τ_{LB} converging to it from below as δ grows (0.7611 / 0.7606 / 0.7606 / 0.7603). The upper endpoint is vacuous once the worst case saturates $q = 1$, which is expected: a bound that allows the nuisance to be maximally favourable carries its information in the lower endpoint.

5.4. What we therefore report

B (unadjusted) as the localization estimand, $B|z$ as the motion-mediated share, both with video-clustered intervals, and the E-value of B as a bound on how much unmeasured confounding (time position, camera motion, annotation error) the claim could survive. We do *not* propose an adjusted estimator as an improvement: our attempts to build one (post-hoc residualisation,

training-time orthogonalisation) both reduce B on real checkpoints, which is what the mediator diagnosis predicts (Section 6.5).

5.5. The pair-weighted estimator

For a frozen scorer and per-video label vectors y_v,

$$\begin{aligned} \hat{B} &= \frac{\sum_v a_v b_v \hat{B}_v}{\sum_v a_v b_v}, \\ \hat{B}_v &= \frac{1}{a_v b_v} \sum_{i:\, y_i=1} \sum_{j:\, y_j=0} \mathbb{1}[s_i > s_j]. \end{aligned} \tag{8}$$

$\hat{B}_v$ is the Mann–Whitney statistic [32] of video v and is computed from ranks in $O(n_v \log n_v)$; the whole estimate costs one sort per video. $\hat{B}$ is the pair-count weighted mean, *not* the unweighted mean of the $\hat{B}_v$ — the two differ systematically, because videos with few positive frames (short events) have both noisy and differently weighted $\hat{B}_v$. On UCF-Crime the difference is not academic: the unweighted mean over videos is 0.7046 for the base checkpoint while the pair-weighted $\hat{B}$ is 0.7558 (0.05 apart), and in a motion baseline the two move in *opposite* directions relative to the model (unweighted 0.7186 vs pair-weighted 0.6552). We report the pair-weighted version because it is the component of θ (Theorem 1); the unweighted version answers a different question ("does the average video localise?") and should be labelled as such.

5.6. The stratified estimator

Motion is observed per unit. For a nuisance z (here: the 5-frame moving average of $\|\Delta x_t\|$, standardised inside the video) define Q-quantile bins inside each video, keep only pairs whose two members fall in the same bin, and average with pair weights. With $Q = 5$ this is a matched comparison: the anomalous and background units being compared have the same motion energy to within a quintile. (Five rather than ten bins is a bias–variance compromise: each bin then keeps enough control units for the within-bin contrast to be estimable in short videos; $Q = 10$ lowers $B|z$ by 0.7 point on the base checkpoint and does not change any conclusion.) Two diagnostics accompany it: the *overlap* (the fraction of positive units that share a bin with at least one control unit; 0.943 on UCF-Crime, per-video mean 0.941) and the same statistic evaluated on z itself, which must fall towards 0.5 if the stratification has taken the nuisance out (it does: $0.6552 \to 0.5465$).

A residualised variant — the natural "de-biased" estimator, ridge-fitted on control units and subtracted out-of-fold — lowers B from 0.7558 to 0.7392 on the base checkpoint, exactly as the mediator diagnosis predicts; we report it as a sensitivity check only, and it is the reason we do not sell it as a method (Appendix Appendix C).

5.7. Inference: the video is the unit

$\hat{B}$ averages over about 10^6 pairs but only 140 anomalous videos, and the frame-level pairs inside a video are strongly dependent. Any interval that treats pairs as independent is off by orders of magnitude. We use a **cluster bootstrap** [33]: draw V videos with replacement, recompute $\hat{B}$ and $B|z$ on the resampled set (videos with no positive or no negative units contribute nothing), and take percentile intervals over 2,000 replicates. For the base UCF-Crime checkpoint this gives $B = 0.7558$ $[0.6844, 0.8364]$ and $B|z = 0.7166$ $[0.6467, 0.7907]$ — a $\pm$ 0.08 interval, i.e. **the between-video variation dominates everything the leaderboard reports**. Reporting B without such an interval would repeat the original sin at a smaller scale.

5.8. An E-value for the contrast

Treat "unit i outscores unit j" as the outcome and the status of i as the exposure. Then $P(\text{win} \mid \text{anomaly}) = \tau$ and $P(\text{win} \mid \text{background}) = 1 - \tau$, so the pair-level risk ratio is $RR = \tau/(1-\tau)$ and the E-value of VanderWeele and Ding [22] applies:

$$E = RR + \sqrt{RR\,(RR - 1)}. \tag{9}$$

E is the minimum strength of association (on the risk-ratio scale) that an unmeasured confounder would need with *both* the status and the score to explain away the contrast. For the base checkpoint $B = 0.7558$ gives $E = 5.6$ and $B|z = 0.7166$ gives $E = 4.5$: to be an artefact, an unmeasured within-video factor would have to raise the odds of being anomalous and of scoring high by more than 4.5-fold each. That is a substantive statement about the evidence, and it is cheap to compute. Two things must be said against it, and the first is arithmetic. An E-value computed at a point estimate is the optimistic end of a bound: computed at the lower end of the video-clustered interval the same checkpoint gives $E \approx 3.8$ for B (0.684) and ≈ 3.1 for $B|z$ (0.647), and those are the values a reader should quote. The second is structural: because $P(\text{win} \mid \text{background}) = 1 - \tau$, the risk ratio is a

deterministic function of τ, so E is a monotone relabelling of B — it supplies a scale that a reader can argue about, not information B does not already contain, and it inherits every construct-validity question B has [34]. Caveat: the E-value also treats pairs as independent; with clustering it is optimistic, so we report it alongside, not instead of, the bootstrap interval.

5.9. Protocol

For every reported system, publish the four numbers θ, B, $B|z$ and E, the last three with a video-clustered 95% interval.

The cost is one extra evaluation pass on frozen scores (seconds) and one paragraph of text. The rest of the protocol follows from the theorems:

1. Compute the pair census once per split and state ω (it is a property of the data, not of the method).
2. Do not claim localization from θ. By Theorem 3 a $\Delta\theta$ is compatible with any ΔB; localization claims must cite B or $B|z$. Cite them as a diagnostic column, not as a replacement headline number: by Section 6.7 we could not certify B against detection mAP, and we will not ask others to report a number whose external criterion we cannot name.
3. Require the interval. A B difference on UCF-Crime is not resolvable at the resolution the field currently quotes: the between-seed sd is 0.018 on B and 0.027 on $B|z$, the cluster sd of a single checkpoint is 0.037–0.053, and the largest gap inside our eleven-arm family ($\Delta B = 0.0495$) has an interval that includes zero (Section 6.6). Same-seed comparisons, whose differences are paired and therefore partly correlated, resolve differences of roughly 0.02. Numbers below that are decoration.
4. Report $D = B - B|z$ as the motion-mediated share, and $B|z$ itself when the claim is about *non-motion* evidence.
5. Treat $\mathcal{A}$-only gains as scene-level: by Theorem 2 they can be obtained with a scorer that is constant inside each video.
6. Quote the bound, not the point estimate alone: state $\delta^* = 2B - 1$, the nuisance share below which the localization claim survives (Section 5.3).

A one-command implementation. The protocol costs one evaluation pass over a frozen score file — no model, no retraining, no GPU. For the base checkpoint, with 2,000 bootstrap replicates and $Q = 5$, the report is: $\omega_A = 0.6317$ and $\omega_B = 0.00388$, i.e. an exchange rate of 257.7; $\theta = 0.8772$ with a video-clustered interval $[0.8237, 0.9152]$; $B = 0.7558$ $[0.6844, 0.8364]$;

$B|z = 0.7166$ $[0.6467, 0.7907]$; $D = 0.0392$ with mediation share $\pi_{\text{mot}} = 0.138$; $E(B) = 5.64$ and $E(B|z) = 4.49$; $\delta^* = 0.512$, with $\tau \in [0.6948, 0.9448]$ at 20% contamination; and a within-video contribution of 0.00294 to θ (0.33%). The same evaluation answers the reverse question, which is the one a localization claim implicitly asks: it converts a headline improvement $\Delta\theta$ into the movement it demands of $\Delta B = \Delta\theta/\omega_B$ and compares that with the remaining headroom $1 - B$ and with the split's own seed noise. On UCF-Crime a claim of +0.8 points of AUC demands $\Delta B = 2.06$, larger than the 0.24 of headroom that exists, so no scorer can support that claim on this split; the arithmetic follows from $\omega_B = 0.00388$ and from nothing else.

6. Experiments

6.1. Host model, checkpoints, and verification

We use DSANet [12], a current state-of-the-art weakly-supervised detector that operates on frozen CLIP ViT-B/16 features with a disentangled semantic-alignment head. Of the fourteen UCF-Crime checkpoints stored on the training machine, eleven are used here; the other three are excluded on reconstruction grounds (two could not be rebuilt from their stored configuration, and one rebuilds to a different pooled AUC). All four XD-Violence checkpoints are used. One of the excluded checkpoints (a motion-injection variant) rebuilds to $\theta = 0.8494$ against an archived 0.8779; we mention it because the discrepancy — six times the between-seed noise — is exactly the kind of silent reconstruction error that a third-decimal headline number invites.

Verification. For every retained checkpoint we recomputed θ from the stored frame scores and compared it with the number archived by the training run: agreement was $< 10^{-4}$ in all cases (0.8772/0.8772, 0.8800/0.8800, 0.8717/0.8717, ...). The decomposition of Theorem 1 reproduced θ to $< 10^{-12}$. For XD-Violence we ran the published test path (with the XD prompt set) and reproduced the repository's printed pooled AUC exactly (0.9449 base-s789, 0.9437 base-s555), which validates the frame-to-snippet alignment that the pair census depends on.

Controlled probes. Because the published checkpoints are few and their architecture is fixed, we also train a deliberately minimal family — a linear frame head with top-k MIL on the *same cached features* (4,000 training clips, 40–60 epochs, 3 seeds) — which lets us vary one design decision at a time and

Table 2: Exact three-block decomposition of frame-level micro-AUC on UCF-Crime, for the eleven checkpoints that survived our re-evaluation. A, B, C are the block statistics, $B|z$ the motion-stratified variant, corr the mean within-video correlation between score and motion energy. Ranks are within the eleven rows.

#	checkpoint	θ	A	B	$B\|z$	C	corr	rank θ	rank $B\|z$
1	base s234	0.8800	0.9751	0.7270	0.6729	0.7167	0.150	1	6
2	motion6 s789	0.8778	0.9728	0.7576	0.7230	0.7144	0.233	2	1
3	base s789	0.8772	0.9773	0.7558	0.7166	0.7050	0.141	3	2
4	cons0 s234	0.8759	0.9774	0.7235	0.6744	0.7016	0.145	4	5
5	attr s234	0.8724	0.9788	0.7080	0.6545	0.6899	0.172	5	11
6	core14txt s234	0.8717	0.9725	0.7169	0.6728	0.6989	0.151	6	7
7	motion6 s555	0.8717	0.9728	0.7180	0.6604	0.6981	0.288	7	10
8	base s555	0.8705	0.9752	0.7236	0.6688	0.6907	0.139	8	8
9	core8g1 s234	0.8684	0.9759	0.7124	0.6621	0.6837	0.126	9	9
10	res s234	0.8682	0.9767	0.7467	0.7093	0.6814	0.328	10	3
11	motion6-shuf s234	0.8616	0.9670	0.7171	0.6786	0.6806	0.121	11	4
	spread	0.0184	0.0118	0.0496	0.0685	0.0361	–	–	–

gives the arms of Section 6.3 and Section 6.5. These probes are not meant to be competitive; they are meant to be interpretable.

6.2. The decomposition on published checkpoints

Four facts about Table 2 carry the paper.

(a) The ω_B share is an invariant of the split, not of the model. The contribution of B to θ is $\omega_B B$: for every row it lies between 0.0027 and 0.0029, i.e. **0.31–0.33% of the reported number**. The remaining 99.68% is scene-level (A, C). A corollary worth stating in words: the architecture that wins the UCF-Crime leaderboard wins it on the 99.7% of the metric that a video-level classifier could also produce.

(b) θ does not rank localization. $\rho_s(\theta, B|z) = 0.218$, 95% CI $[-0.44, 0.72]$; $\rho_s(\theta, B) = 0.418$, CI $[-0.24, 0.81]$; and even $\rho_s(\theta, A) = 0.191$, CI $[-0.46, 0.71]$, so within this architecture family the θ ordering is not explained by the scene-level term either. One sensitivity belongs here, because a rank correlation over eleven points is easily moved by a single arm: the placebo of (d) is a constructed outlier, and recomputing without it gives $\rho_s(\theta, B|z) = 0.239$ and $\rho_s(\theta, B) = 0.294$ at $n = 10$ — the first rises, the second falls, and neither reaches significance. The intervals are wide — eleven points cannot exclude a moderate positive association, so we read

these correlations as *inconclusive rather than as evidence of independence*, and we rest the claim on the decomposition instead: the localization term is 0.31–0.33% of θ by construction, which is a statement about the metric and not about eleven draws. Meanwhile the two localization measures agree with each other: $\rho_s(B, B|z) = 0.800$ (CI $[0.39, 0.95]$, $p = 0.003$). The leaderboard and the localization ranking are not the same ranking.

(c) The localization spread is large — and invisible. $B|z$ spans 6.9 points (0.6545 to 0.7230), four times the θ spread (1.8 points). At the exchange rate that entire spread is worth $0.00388 \times 0.0685 = 0.00027$ AUC — one eighteenth of the seed-to-seed noise (Section 6.4). The community's practice of comparing θ to the third decimal therefore cannot see a difference that is, in localization terms, an order of magnitude larger than anything it does see.

(d) A placebo arm is not last. The eleventh row is a deliberately corrupted model: the motion channels were computed and then shuffled during training, so the feature is noise. It has the *lowest* θ (0.8616) — as intended, it is visibly broken — but it ranks **4th of 11** by $B|z$ (0.6786), ahead of six honest arms; by the unadjusted B the same arm ranks 8th, so it is the stratified column that lifts it. Whatever $B|z$ is measuring, it is not "model quality" in the sense the leaderboard means; and any audit that promoted it would be promoting a corrupted run. This is a caution about single-number localization statistics (including ours) that we take seriously in Section 6.8 and Section 7.6.

6.3. Degeneracy in practice

The constant scorer keeps **99.7% of the identically-trained frame-level model's** θ (0.8234 vs 0.8257) and **93.9% of the published DSANet checkpoint** (92.1% of the best number reported in the literature), while its localization term is *exactly* chance (B = 0.5000, all within-video pairs tied). Note also that the constant scorer has a *higher* A than the frame model (0.9766 vs 0.9571).

Theorem 2 predicts the shape of this table, and it also says how to read the 0.0023 that separates the two scorers — which is *not* the price of localization. Term by term, $\omega_A \Delta A = 0.6317 \times (-0.0195) = -0.0123$, $\omega_B \Delta B = 0.00388 \times (+0.1571) = +0.0006$ and $\omega_C \Delta C = 0.3645 \times (+0.0385) = +0.0140$: the localization term contributes six tenths of a thousandth, and the headline difference is the near cancellation of a twelve-point loss in the scene term against a fourteen-point gain in the severity term (the implied C is 0.561

Table 3: Constant-within-video scorers against frame-level scorers on the same cached CLIP features and training protocol (UCF-Crime, linear heads, top-k MIL for the frame models). The last column is the *unweighted* mean of the per-video within-video AUC, which coincides with the pair-weighted B for the first three rows and differs for DSANet, whose videos carry unequal numbers of pairs.

scorer	θ	A	B	within-video AUC
video-level classifier (one score per video)	0.8234	0.9766	0.5000	0.5000
per-video maximum of frame scores	0.7959	–	0.5000	0.5000
frame-level MIL (same features, protocol)	0.8257	0.9571	0.6571	0.6571
DSANet base s789 (published)	0.8772	0.9773	0.7558	0.7046

for the video-level scorer and 0.600 for the frame model). The frame model is a better localizer by 15.7 points of B and is still separated from a scorer that cannot localize at all by 0.0023 of θ — half of one seed standard deviation, and far below the third-decimal reporting granularity the community uses to claim progress. What the 0.0023 measures is a cancellation, not the value of localization.

6.4. Resolution: what the metric can and cannot represent

Three base checkpoints that differ only in seed — rows 1, 3 and 8 of Table 2 — give $\theta = 0.876 \pm 0.005$, $B = 0.736 \pm 0.018$ and $B|z = 0.686 \pm 0.027$ (mean $\pm$ one standard deviation over the three runs). Localization is therefore three to six times as noisy as the headline number, and that noise is invisible in it.

Two consequences, both quantitative.

First, by Theorem 3 the standard error of a ΔB imputed from a single-seed $\Delta\theta$ is $0.0049/0.00388 = 1.26$ — 2.5 times the full parameter range of B. Nothing about localization can be read off an AUC comparison at the precision this literature reports.

Second, the *attainable* range: since $\mathrm{B} \in [0.5, 1]$, a localization improvement that went from chance to perfect would move θ by $\omega_B/2 = 0.0019$, which is 0.4 of the seed standard deviation. The entire observed localization spread across eleven checkpoints (Section 6.2) is worth 0.06 seed-σ. **The reported metric is strictly less sensitive to localization than the training run is to its random seed.** This is not a claim about current methods; it is a

property of the split and the statistic.

The same computation gives the planning number for anyone who wants to detect a localization difference with the standard metric: to certify $\Delta B = 0.01$ with a two-standard-error margin you need $\sigma(\Delta\theta) \leq 1.9 \cdot 10^{-5}$ — a standard error 253 times smaller than the observed run-to-run noise, which at $\sigma \propto k^{-1/2}$ would take roughly 64,000 seed replicates and is unattainable in this literature (most papers report one run). Conclusion: measure B.

6.5. Motion behaves as a mediator, not a confounder

We run the three interventions of Section 5.2.

Observational stratification (Table 2, column $B|z$): $D = B - B|z$ is 3.45–5.77 points across the eleven checkpoints, median 4.92. Two normalisations of that gap circulate and they are different numbers, so we fix both here. Relative to the remaining headroom, $\pi_{\text{mot}} = D/(1 - B|z)$, the median is 0.149 (Section 5.3); relative to the distance from chance, $D/(B - 1/2)$, the median is 0.220. It is the second that the sentence "roughly **5 of the 25 points that separate the model's within-video AUC from chance are motion-mediated**" expresses — 4.92 points of a 25-point span — and only the first appears in the tables. A large share either way, but a minority.

Intervention on training. Adding $\gamma\,\text{mean}_v\text{corr}^2(S_v, Z_v)$ where Z is the 5-frame motion energy, on the linear probe:

The placebo is inert — it is statistically indistinguishable from the baseline (0.6891 vs 0.6805 for B; 0.6480 vs 0.6429 for $B|z$) and it does *not* reduce the correlation with real motion (0.169). The degradation under the real penalty is therefore specific to real motion structure rather than to regularisation as such — though we note again that a nuisance permuted across videos is nearly uncorrelated with the score by construction, so the placebo is a weak control and its inertness is close to guaranteed. On the mediator reading — the annotated events are largely motion-defined, so motion lies on the path from anomaly to score — adjusting for it is over-adjustment; on the confounder reading, $B|z$ would be the honest estimand. We report both and return to the limits of this verdict in Section 5.2.

Post-hoc residualisation on real checkpoints. Cross-fitted subtraction of a ridge fit of the score on four nuisance coordinates — the normalised time position t/n_v, the instantaneous motion $\|\Delta x_t\|$, its 5-frame moving average $\overline{\|\Delta x_t\|}$ and the feature norm $\|x_t\|$ — with folds assigned by video parity, on base s789 (intervals: 500 video-cluster bootstrap replicates):

Table 4: The three interventions on the nuisance. Top: an orthogonality penalty against the true motion statistic, three seeds per row, reported as their mean. Bottom: the same penalty at $\gamma = 1$ against real versus shuffled motion. The penalty does decorrelate the score from motion, and it lowers the localization estimate rather than raising it. The $\gamma = 1$ real-motion run is one run and appears in both blocks, so its four numbers are stated identically twice.

condition	θ	B	$B\|z$	corr
training-time orthogonalisation (mean of three seeds)				
$\gamma = 0$ (baseline)	0.8352	0.6805	0.6429	0.163
$\gamma = 0.1$	0.8331	0.6772	0.6417	0.154
$\gamma = 0.3$	0.8300	0.6744	0.6426	0.144
$\gamma = 1.0$	0.7980	0.6236	0.6043	0.096
placebo: $\gamma = 1$ against shuffled motion				
baseline ($\gamma = 0$)	0.8352	0.6805	0.6429	0.163
$\gamma = 1$, real motion	0.7980	0.6236	0.6043	0.096
$\gamma = 1$, shuffled	0.8362	0.6891	0.6480	0.169

Every variant lowers both localization measures on both checkpoints — by 1.7–4.4 points for B and 0.7–1.8 for $B|z$ — which the mediator diagnosis predicts (the confounder reading predicts the same sign, Section 5.2). They also cost 1.0–1.9 points of θ, so a paper selling them as a de-biasing method would be selling a method that lowers the very quantity it claims to improve.

Three independent lines of evidence for the same conclusion. (i) A zero-training motion statistic (5-frame moving average of $\|\Delta x_t\|$) reaches $B = 0.6552$ and $B|z = 0.5465$: 86% of a trained probe's excess over chance is reproduced by motion, and all of it is absorbed by stratifying on motion. That number is the sharpest limit on the construct and we do not soften it: in the only block that measures localization, a training-free motion statistic reproduces most of what a trained frame-level probe reports. It is why the column is described throughout as *within-video ranking concordance* — what B measures by construction — rather than as localization ability, and why the 86% is a statement about the construct rather than a defect of the estimator. (ii) Trained checkpoints retain a within-video score–motion correlation of 0.12–0.33 (mean 0.18 across eleven; Table 2) that cannot be removed without removing the signal. (iii) The one arm with the *highest* score–motion correlation (res s234, 0.328) is also the third best by $B|z$ (0.7093), so

Table 5: Post-hoc residualisation on two checkpoints. The score is regressed on four nuisance coordinates, the fit is removed with video-parity cross-fitting, and the report is recomputed. Every variant lowers both localization measures on both checkpoints. Video-cluster bootstrap intervals (500 replicates) for the raw rows are $B \in [0.684, 0.836]$ and $B|z \in [0.647, 0.791]$ on base s789, and $[0.681, 0.819]$, $[0.656, 0.774]$ on res s234. The E column is the E-value of $B|z$, not of B.

variant	θ	B	$B\|z$	E
base s789 (score–motion correlation 0.141)				
raw	0.8772	0.7558	0.7166	4.50
residualised, all frames	0.8600	0.7356	0.7099	4.33
residualised, control frames	0.8668	0.7392	0.7026	4.16
res s234 (score–motion correlation 0.328)				
raw	0.8682	0.7467	0.7093	4.31
residualised, all frames	0.8493	0.7028	0.7001	–
residualised, control frames	0.8532	0.7031	0.6915	–

the correlation is not monotonically a symptom of "detecting motion instead of anomalies".

6.6. The audit: what happens when the leaderboard is re-sorted

Ranking the eleven checkpoints by $B|z$ instead of θ (Section 6.2, last two columns) changes the conclusion in four ways: (i) the θ-winner (base s234, 0.8800) falls to 6th; (ii) a run ranked 10th by θ (res s234, 0.8682) rises to 3rd; (iii) the checkpoint with the best *scene* term (attr s234, A = 0.9788, rank 5 by θ) is last by localization (11th of 11); and (iv) the placebo arm ranks 4th.

The rank correlation is 0.22 with a 95% interval that includes zero ([-0.44, 0.72], p = 0.52). Eleven checkpoints cannot demonstrate independence, and we do not claim they do: the audit shows that the ordering by θ is not an ordering by the localization term *of the same metric* (0.31–0.33% of it), while the rank evidence alone is merely inconclusive — on this family of models, which is the only family we can compare *architectures* within at fixed features and protocol.

The audit is also where we owe the reader the sharpest limit of the paper, so we state it here rather than in a footnote. A paired cluster bootstrap over videos (the protocol of Section 6.4, 2,000 resamples, the *same* resample applied to every arm so that the differences are paired) gives a cluster standard deviation of 0.037–0.053 on B and 0.031–0.048 on $B|z$ for a single checkpoint.

The largest between-arm gap that the family contains — motion6 s789 against attr s234, $B = 0.7576$ vs 0.7080 — is $\Delta B = 0.0495$ with a 95% interval of $[-0.005, +0.095]$: it includes zero. The top two arms by B swap order in 51.2% of resamples (51.3% by $B|z$), i.e. the ordering of the best checkpoints is a coin toss at this sample size, and arms that differ only in seed partially share their variance (a paired ΔB between neighbouring arms has a standard deviation near 0.011) without becoming resolvable.

So the audit should be read as a statement about *sets*, not about ranks: the θ ordering does not carry the localization ordering, but neither does the B ordering carry a certification that one arm localizes better than another by four decimals. What the eleven-arm family supports is (i) the decomposition and its weights, (ii) the *magnitude* of the gap between the two rankings, and (iii) the demonstration that a re-sort is possible at all; it does not support a new leaderboard of its own, and we do not offer one. This is exactly why the recommendation of Section 5.9 is a third column with an interval, not a new headline. Adding the released VadCLIP weights as a twelfth row changes none of this: the gap, its interval and the flip rate are unchanged ($\Delta B = 0.0495$ over $[-0.005, +0.095]$, 51.4% flips), and the released weights sit tenth of twelve by B — inside the band, below its top.

We are careful about what this does and does not show. It does not show that the published ordering is wrong: A and C are legitimate quantities (a detector that separates anomalous from normal *scenes* does something useful, and C measures severity ranking). It shows that the ordering is not an ordering *by localization*, while the papers read it as one. The remedy is not to replace the metric but to report the missing column.

6.7. Criterion validity: do the extra columns predict detection mAP?

A re-sorting argument is only as good as its localization columns, and a reviewer is entitled to ask what B predicts that anybody outside this paper cares about. The natural outside criterion is the detection metric that the same papers report, so we compute two of them, on every checkpoint and from the same per-frame scores that produce θ and B, rather than quoting numbers from training logs. The first is TLoC-style detection mAP at IoU 0.5 under the evaluation routine the UCF-Crime detection literature shares (segment mAP, excludeNormal = False, IoU 0.1–0.5, fourteen classes, per-class proposals from a relative score threshold, 0.7× class score added to the proposal score, NMS at 0.6). We run that reference code as published and, in the same process, an independent re-derivation of the same algorithm written for this

paper (vectorised run extraction, its own non-maximum suppression, its own greedy matcher). The two agree exactly: the largest absolute deviation over all 5×11 cells is 0.0. This column is therefore the published protocol and not our variant of it. The second implementation is still reported, because it does not inherit the reference code's per-class score gating and is short enough to check by hand: nine relative thresholds per video, maximal runs above each threshold as proposals, a proposal's confidence taken as the mean score inside the run, greedy matching at IoU 0.5 and an interpolated precision-recall integral.

The detection numbers are small in absolute terms, and the reason is a definition rather than a defect. The papers on this benchmark publish frame-level AUC and average precision, not detection mAP: for the released checkpoint we reproduce the frame-level AUC its own paper reports on the same test split (0.8808 against 88.02), so the scores are the model's and the pipeline that turns them into numbers is calibrated at that level. Under the detection protocol the same checkpoint scores 2.95 mAP@0.5, and our eleven arms span 2.44–5.82; that metric credits one segment per video above everything else, and weakly supervised models trained on frame-level labels are not optimised for it. We therefore use the detection columns as *rank* criteria inside one fixed protocol, never as absolute numbers comparable to the literature, and we rest part of the reading on the agreement of the two implementations: $\rho_s = 0.57$ between them across twelve checkpoints. We also correlate against the frame-level average precision itself, since that is the number these papers do report: it lies between 0.294 and 0.374 across the eleven self-trained checkpoints and is 0.339 for the released anchor.

What the recomputation does not buy is a second measurement. The two detection implementations share our scores and our labels, and $\rho_s = 0.57$ is weak agreement between two pieces of code, not two experiments. Nor is the ground truth better than it is: the file the protocol reads stores segments at 16-frame granularity and marks the 150 normal test videos with a placeholder class label, so any proposal inside them is counted as a false positive by construction. What survives all of that is a rank comparison inside one fixed protocol, which is what the next paragraph reports. Nothing here establishes that detection mAP *is* the right criterion for B, and no external criterion in this paper does.

Three correlations settle the question for each metric, all on twelve points, all reported with the Fisher-z interval that twelve points can support, and all with a two-sided permutation p over 10^5 sampled orderings of the crite-

Table 6: Criterion validity: the reported number, the two localization columns, and two independently implemented detection metrics, for the eleven self-trained checkpoints and the released VadCLIP weights (last row, marked †). All three metric columns are percentages produced under one fixed protocol each and are used as rank criteria only. The mAP@0.5 and avg mAP columns are the benchmark's reference protocol, whose values an independent re-derivation reproduces exactly; the last column is our own shorter implementation with a different proposal rule.

checkpoint	θ	B	$B\|z$	mAP@0.5	avg mAP	AP@0.5
base s234	0.8800	0.7270	0.6729	5.68	10.10	4.46
motion6 s789	0.8778	0.7576	0.7230	4.61	12.46	2.41
base s789	0.8772	0.7558	0.7166	5.18	12.84	3.94
cons0 s234	0.8759	0.7235	0.6744	5.46	11.16	2.51
attr s234	0.8724	0.7080	0.6545	5.82	13.03	2.79
core14txt s234	0.8717	0.7169	0.6728	5.03	11.88	4.24
motion6 s555	0.8717	0.7180	0.6604	2.88	10.15	2.86
base s555	0.8705	0.7236	0.6688	4.12	11.30	2.53
core8g1 s234	0.8684	0.7124	0.6621	4.05	10.64	2.45
res s234	0.8682	0.7467	0.7093	4.63	9.83	1.89
motion6-shuf s234	0.8616	0.7171	0.6786	2.44	7.17	1.85
VadCLIP†	0.8560	0.7157	0.6713	2.95	7.36	1.18

rion (Table 6). For the detection protocol, $\rho_s(\theta, \text{mAP@0.5}) = 0.71$, 95% CI $[0.24, 0.91]$, $p = 0.012$; $\rho_s(B, \text{mAP@0.5}) = 0.15$, CI $[-0.47, 0.66]$, $p = 0.65$; $\rho_s(B|z, \text{mAP@0.5}) = 0.08$, CI $[-0.52, 0.62]$, $p = 0.82$. The independent implementation returns the same pattern with different numbers: 0.69 $[0.20, 0.91]$ for θ ($p = 0.011$), 0.11 $[-0.49, 0.64]$ for B ($p = 0.74$) and -0.20 $[-0.70, 0.42]$ for $B|z$ ($p = 0.54$); the average-mAP column agrees again (0.63 $[0.09, 0.88]$ for θ, 0.10 for B).

The frame-level average precision behaves differently, and we report the difference rather than the half of it that suits the argument. On all twelve points it tracks neither column ($\rho_s(\theta, \cdot) = 0.45$, $p = 0.14$; $\rho_s(B, \cdot) = 0.33$, CI $[-0.30, 0.76]$, $p = 0.29$), but that is the released anchor's doing: its average precision is unremarkable (0.339) while its θ is the lowest in the table. Drop the anchor and the eleven self-trained points give $\rho_s(\theta, \cdot) = 0.74$, CI $[0.24, 0.93]$, $p = 0.013$, and $\rho_s(B, \cdot) = 0.49$, CI $[-0.15, 0.84]$, $p = 0.13$ — an interval that admits zero, a moderate association and a strong one, so this criterion neither certifies B nor excludes it.

Two cautions on the partial correlations, which we computed and do not rest on. Partialling θ out conditions on a variable that contains B

($\theta = \omega_A A + \omega_B B + \omega_C C$), which mechanically biases the partial association downward; and at $n = 12$ no partial association here is distinguishable from zero (-0.32, permutation $p = 0.31$, under the detection protocol; $+0.14$, $p = 0.66$, for the frame-level criterion). The clean partialling — on A and C, the other two blocks — is the analysis a follow-up should run.

The honest reading is that mAP, as we can measure it, does not supply the missing column either. It tracks θ — which is what the decomposition predicts, since 99.7% of the pairs driving any frame-level score are cross-video pairs and segment AP over 16× repeated frame scores is dominated by them — and it shows no association with B beyond what θ already carries, with intervals wide enough that a moderate association is not excluded. We did not expect this when we designed the check; it is a negative result for the most obvious way to validate B, and we report it as one. Three things follow, and they constrain the claims of this paper.

1. B is not offered as a replacement headline number. We cannot certify it against the field's own localization metric, so promoting it to the leaderboard would be a stronger claim than these data support.
2. Neither field metric, as we can measure it, orders the axis the theorems say is missing. Frame-level AUC sees 0.39% of within-video pairs; detection mAP, which one would expect to see them, tracks θ instead; and the frame-level average precision, the number the papers publish, leaves the question open at this sample size rather than answering it. Certifying or refuting within-video ordering against an outside criterion remains open.
3. The remedy remains reporting, not replacing: three columns instead of two, with the third one explicitly labelled as a diagnostic whose external validation is an open problem (Section 7.6).

The twelfth row is an external anchor. The released VadCLIP weights, scored by this paper's suite rather than by its own authors, reproduce the published frame-level AUC under the repository's own 16× convention (0.8808 against the 0.8802 the paper reports) and land inside the band our eleven arms span on the snippet grid used throughout ($B \in [0.708, 0.758]$, $B|z \in [0.655, 0.723]$). The two AUC conventions, the unit-of-analysis difference, and the mAP comparison are detailed in Appendix Appendix D.

One numerical caution belongs here rather than in a footnote. Twelve checkpoints, a mAP range of 1.2–5.8% and a between-seed mAP spread comparable to the between-architecture spread mean that these rank correlations

are fragile: the intervals above are wide by construction, only the θ association survives a two-sided test, and nothing in this table should be read to the second decimal.

6.8. Inference in practice

Clustered intervals. The bootstrap of Section 5.7 (2,000 replicates, videos as clusters) gives $B = 0.7558$ $[0.6844, 0.8364]$ and $B|z = 0.7166$ $[0.6467, 0.7907]$ for the base checkpoint: a $\pm$ 0.07 window for a quantity estimated from 140 anomalous videos.

The same exercise is instructive for the *reported* number. On identical scores, a frame-level i.i.d. bootstrap — the assumption behind a DeLong-style interval [35] — gives $\theta = 0.8772$ $[0.8734, 0.8808]$, a width of 0.0075, while resampling *videos* gives $[0.8237, 0.9152]$, a width of 0.0916: **12.3 times wider**. The precision the field attributes to its headline number is therefore partly an artefact of treating 69,368 frames as 69,368 independent observations when they are 290. This is a milder version of the error this paper is about, and it has the same remedy: state the cluster. ($B|z$'s clustered interval, 0.144, is in fact *wider* than θ's 0.092: conditioning on motion keeps only same-bin pairs, and the discarded cross-bin pairs are the honest price of the stratified estimator.)

Between-seed noise. $\mathrm{sd}(B|z) = 0.0267$ across three seeds of the same configuration, i.e. two fifths of the full 0.07 spread of Table 2. Any single-seed localization claim below about 0.05 on UCF-Crime is not established, and papers that report one run should say so.

Overlap. 94.3% of positive units share a motion quintile with at least one control unit in the same video (per-video mean 0.941), so the stratified estimator is well supported; the remaining 5.7% come from videos whose anomalous segment is uniformly more dynamic than its background — exactly the cases where the motion question is unresolved.

E-values. $E(B) = 5.6$ and $E(B|z) = 4.5$ for the base checkpoint: an unmeasured within-video confounder would need to be associated with both the status and the score by a risk ratio of 4.5 to explain the localization contrast away. Given that we have measured and ruled out the dominant candidate (motion), this is a defensible statement of robustness — and the number to report instead of the words "we controlled for scene bias".

Table 7: The decomposition on XD-Violence (800 videos, 2,331,296 scored frames, 537,882 positive frames). The pooled number agrees with the repository's own frame-level AUC to six decimals.

checkpoint	θ	A	B	$B\|z$	C	corr	rank θ	rank $B\|z$
base s234	0.9536	0.9966	0.8466	0.8223	0.8540	0.192	1	1
base s789	0.9449	0.9962	0.8281	0.8000	0.8258	0.200	2	3
base s555	0.9437	0.9969	0.8419	0.8160	0.8204	0.203	3	2
motion3 s789	0.9368	0.9920	0.8046	0.7774	0.8090	0.201	4	4

6.9. XD-Violence

XD-Violence is the harder test of the framework: it has an order of magnitude more frames, denser annotation (537,882 positive frames of 2,331,296), six violence categories, and the field reports it with AP/mAP over IoU thresholds rather than micro-AUC. Four DSANet checkpoints were available (three seeds of the base configuration and one motion-injection variant).

The weights are $\omega = (0.6985, 0.00071, 0.3008)$: **0.07% of the metric's pairs are within-video**, and the localization share of θ is $\omega_B B = 0.0006$, i.e. *less* than on UCF-Crime even though XD-Violence annotates three times as large a fraction of its frames. The reason is arithmetic rather than conceptual: with 800 videos and 1.25M frames in normal videos, the number of cross-video pairs grows much faster than the within-video ones. A benchmark does not become localization-sensitive by annotating more frames; it becomes so by having fewer or shorter videos per split. This is a design lesson that costs nothing to apply and that we make concrete in Section 7.

Two differences from UCF-Crime are worth recording honestly. First, B itself is much higher (0.80–0.85 vs 0.71–0.76): XD-Violence is dense — violence occupies large portions of most anomalous videos — so a within-video contrast has more signal to find, and the localization picture on XD-Violence is genuinely better than on UCF-Crime. Second, on these four checkpoints the θ and $B|z$ orderings largely agree ($\rho_s = 0.8$, p = 0.2, n = 4), unlike the eleven-checkpoint UCF-Crime family of Section 6.2. With n = 4 we draw no conclusion from either; we note that the two settings differ in what they vary (design vs seed), and that our central claims — the ω weights, the exchange rate, the degeneracy, the irreversibility — are properties of the split and the statistic, not of the ordering, so they transfer unchanged.

The motion-mediated share is smaller on XD-Violence ($D = B - B|z$ between 2.4 and 2.9 points, vs 3.5–5.8 on UCF-Crime) while the score–motion

correlation is similar (0.19–0.20). Motion matters, again, but is not the whole story: after stratifying on motion the contrast is still 28–32 points above chance, so on XD-Violence most of the within-video separation is *not* motion-mediated. On UCF-Crime the corresponding figure is 15–22 points, of which motion accounts for roughly a quarter.

The seed spread is comparable to UCF-Crime ($\mathrm{sd}(\theta) = 0.0055$, $\mathrm{sd}(B) = 0.0095$, $\mathrm{sd}(B|z) = 0.0114$ over three seeds), so the resolution argument of Section 6.4 holds with the same arithmetic: the localization limits are $\omega_B/2 = 0.00036$ for motion-independent perfect localization, and one reported XD point is worth 1,408 within-video points.

6.10. Summary of the empirical claims

The empirical content of the paper reduces to eight numbers, each checkable from the released scores: the within-video pair share ($\omega_B = 0.00388$ UCF / 0.00071 XD); the 99.7%/93.9% degeneracy figures; the 6.9-point localization spread worth 0.00027 AUC against seed noise of 0.0049; the inconclusive $\rho_s(\theta, B|z) = 0.22$; the mAP criterion failure ($\rho_s = 0.71$ with θ against 0.15 with B); the 5-of-25-point motion share; the nuisance-share tolerance $2B - 1 = 0.42$–0.52 (UCF) / 0.61–0.69 (XD); and the unresolvability of the eleven-arm spread as a ranking (Section 6.6). Section 6.7 is the one we expected to come out differently.

7. Discussion

7.1. The answer in one sentence, and what follows from it

Theorems 1–4, Proposition 5 and Tables 2–7 give a single sentence answer: *frame-level micro-AUC is a scene-level severity ranking with a small localization component attached*, and the size of that component is computable in advance from the data. On UCF-Crime it is 0.39% of the metric's pairs and 0.31–0.33% of its value; on XD-Violence 0.07% and 0.06%. Everything else — the 150 entirely normal test videos (51.7% of the split), the ranking of anomalous videos by severity, the separation of scenes — is measured by the cross-video blocks A and C.

This reframes several familiar results. A model that "improves AUC by 1 point" has, at most, moved the localization term by 1/258 of a point (UCF) or 1/1408 (XD), and in practice has moved it by nothing measurable (Section 6.4). A model that loses 1 point of AUC may have *gained* localization (res s234 in Table 2 does exactly this: rank 10 by θ, rank 3 by $B|z$). And

the widely observed fact that video-level classifiers score highly on these benchmarks is not a bug in those classifiers — it is the metric doing what its weights say.

None of this makes the metric useless. For *triage* — deciding which video to send to a human — a scene-level severity ranking is close to the right quantity, and A is a defensible measure of it. The error is one of naming: the field reports a triage metric and reads it as a localization metric.

One claim is withdrawn here rather than in a footnote. It would be convenient if the localization column were also a *better* number to optimize, and Section 6.7 shows that the obvious route to that claim fails: detection mAP, the field's own localization metric, tracks θ and not B. So the decomposition buys a *diagnosis* — what the reported number is made of — and not a replacement. The difference matters for how this work should be used: to read a leaderboard correctly, not to win one.

7.2. Designing a split that can measure localization

The weights are computable before a single model is trained. With n_v units in video v and N_{pos}, N_{neg} the numbers of positive and of negative units in the split, $\omega_B \approx \sum_v a_v b_v / (N_{\text{pos}} N_{\text{neg}})$, so a split measures localization when its anomalous videos are *short* and *few* relative to the number of background frames. XD-Violence annotates 23% of its frames and still gets $\omega_B = 0.07\%$, because it has 800 videos and 1.25M frames in normal videos; annotation density is not the lever, video count is.

Three practical consequences.

1. **Report the census.** One line per benchmark: $\omega_A, \omega_B, \omega_C$ and the exchange rate. It takes seconds and it tells every reader how much of the reported number can possibly be about localization.
2. **Report the exchange rate next to every gain in θ.** On UCF-Crime a claimed gain of 0.9 points of AUC cannot be a localization claim: converted at the exchange rate it demands $\Delta B = 0.009/0.00388 = 2.3$ — more than four times the full range of B (0.5) and nearly ten times the remaining headroom $1 - B \approx 0.24$ — while the whole attainable localization range is worth only 0.0019 of θ, a fifth of the claimed gain. A paper making such a claim should say which block moved.
3. **If localization is the claim, report B.** No new statistic needs to be invented: B is simply micro-AUC with the pair population reweighted to the within-video pairs, and it is the estimand of Theorem 1's second term.

A benchmark that wanted to be localization-sensitive could reweight once and be done — at the cost of a wider interval (Section 6.8), which is the honest price.

7.3. What our own failures say about method research

Every intervention we tried in order to *raise* the localization estimate failed: motion orthogonalisation at training time (γ scan, monotone degradation), cross-fitted residualisation at test time (-1.7 to -4.4 points of B on the two checkpoints we tried it on), and — in a companion study — thirteen post-hoc debiasing variants including an oracle that uses ground-truth distances (-0.0045 AP). Theorem 4 explains the pattern: the weak label constrains a symmetric function of the scores inside an anomalous video, so the *ordering* of most frames within a video is unconstrained by the objective, and no amount of post-processing of a trained scorer's outputs can create information the objective never supplied.

This suggests where method work should go, and where it cannot go. It cannot go into loss reweighting, score calibration, or threshold tuning: those operate on a quantity the label does not identify. It can go into *new evidence* — object tracks, audio, interactions between people, multi-scale temporal semantics — that creates within-video variation the label *can* constrain, because such evidence makes the frame-level ordering identifiable through the top-k pool. That is a research programme, not a loss term, and it is the one direction our negative results leave open.

7.4. Relation to the 2026 audit literature

Two concurrent audits reach compatible empirical conclusions. Song and Lee [1] observe that within-video pairs are 0.071–0.388% of all anomalous–normal pairs and that video-level classifiers can score 81.40–97.18% micro-AUROC with within-video AUC exactly 50%. Abdulaziz and Bondarev [2] show that pooled AUC does not reliably predict within-video localization and that normal clips alone can separate videos by acquisition attributes. Our Sections 3 and 4 supply the algebra those observations are instances of: the pair-count weights (Theorem 1) give the 0.071–0.388% directly from the data, the constant-scorer construction (Theorem 2) gives the exact 50%, and the video-level classifier built on it attains 0.93 of the state of the art on UCF-Crime, and the inverse problem (Theorem 3) quantifies why a pooled difference cannot be read as a localization difference. What we add beyond the audits is the exchange rate, the identification analysis (what B *is*, and

why motion is a mediator rather than a confounder), the estimators with clustered inference, and the reporting protocol.

7.5. Concurrent work that proposes replacement metrics

A third line of work replaces the metric rather than decomposing it. Liu et al. [36] identify three limitations — single-annotation bias, no reward for early detection, and no way to measure scene overfitting — and respond with probabilistic AUC/AP over multi-round annotations, a latency-aware AP, and two hard-normal benchmarks evaluated on ten systems. The difference in kind is worth stating plainly: their remedies require *new annotation rounds and new data*, and they change what is being measured; ours requires *neither*, because the identity in Theorem 1 is a property of the metric the field already reports. The two are complementary and can be combined — a latency-aware or probabilistic AP would be decomposed by the same pair-count argument, and the resulting ω_B would be computed once from the annotation rounds. One caution from Section 6.7 applies to any such proposal, including ours: a metric that is not itself an annotation-round construct must be validated against an outside criterion, and when we tried that for B the outside criterion turned out to be a re-measurement of θ. We therefore read this line of work as evidence that the field accepts metric-level contributions, not as competition for the same slot: they ask what we *should* report, we ask what the reported number *already contains*.

7.6. Limitations and threats to validity

Scope of the evidence: one architecture family, plus one anchor. The decomposition is exact and the weights are data properties, so Theorems 1–3 transfer to any split. The *empirical* claims — the ranking disagreement of Section 6.2, the size of the motion share, the seed noise — are measured on DSANet checkpoints on UCF-Crime and XD-Violence, plus the released VadCLIP weights as a sixteenth checkpoint used as an external anchor rather than as a member of the family (Section 6.7). The machine used for these experiments has no outbound network access, so RTFM and other public releases could not be fetched; the family (VadCLIP features, DSANet host, one protocol) remains a single-model-family study — eleven arms in which the design is fixed while the architecture varies, which is exactly what the decomposition needs, but it is not the published leaderboard, and one anchor does not turn it into one. On XD-Violence the four checkpoints we have agree in their θ and $B|z$ orderings, so the "leaderboard does not rank localization"

finding should be read as established for the UCF-Crime eleven-checkpoint family (which spans design variants and includes a corrupted control) and *not* as a universal law; the metric-level claims do not depend on it. Even inside the family the spread is not resolvable *as a ranking*: the largest gap has an interval that includes zero and the top-two ordering flips in half the resamples (Section 6.6). Re-scoring the remaining published checkpoints would be a straightforward experiment for a group that can reach them; it is not one we can run here, and we prefer to say so than to generalise from eleven arms.

Strained assumptions: annotation, unit scores, and the choice of nuisance. A4 fails to some degree in both benchmarks: annotations are segment-level and coarse, and a companion study found that re-ordering frames by distance to the annotated segment *reduces* AP, i.e. that part of what looks like "halo" error is genuine anomaly outside the label. Under A4 failure, true anomalies are counted as controls, so B is biased *downward* and our estimates are conservative; we do not attempt to correct for it, since doing so requires a model of the annotation process that no benchmark currently supplies. A1 fails for any model whose frame score is produced by attention across the video, which is all transformer-based detectors including our host: B is then a statement about the scorer as a black box evaluated frame-wise, which is what deployment does, but a claim about the *representation* would need an intervention on the model's internal aggregation that we do not attempt. Finally, $B|z$ conditions on motion only; camera motion, lighting changes, and crowd density are plausible within-video nuisances we do not measure. Two partial answers: the bound of Section 5.3 does not require the nuisance to be named at all — it charges an arbitrary share δ of the pair mass and reports the interval — and the E-value gives the confounding a claim could survive. Both are bounds, and the E-value treats pairs as independent and is therefore optimistic relative to the clustered interval.

No external criterion for B. Section 6.7 is the limitation we would most like to have answered and could not. Detection mAP at IoU 0.5, the field's own localization metric, tracks θ ($\rho_s = 0.71$, permutation $p = 0.012$) and not B ($\rho_s = 0.15$, $p = 0.65$) under either of two implementations, so B has no established external criterion on these benchmarks. The frame-level average precision, the number these papers actually publish, is the one criterion that shows a positive association with B ($\rho_s = 0.49$ over the eleven self-trained checkpoints), but with a 95% interval of $[-0.15, 0.84]$ and $n = 11$ that association is not separable from zero, and it weakens to 0.33 once the released anchor enters. Two things limit how far even the negative

reading goes: the detection protocol is faithful — an independent re-derivation matches it exactly — but not comparable to published magnitudes, since the released weights score 2.95 mAP@0.5 under it while the papers report frame-level numbers, which we do reproduce (0.8808 against 88.02); so all we have is a rank criterion, and a rank criterion over twelve points supports only the negative statement that B does not order detection mAP. We therefore present B as a *diagnostic of a construct the current evaluation system does not report*, not as a validated replacement, and we flag the missing criterion as the first thing a follow-up should build: a split with enough within-video pairs to make segment-level localization a dominant term, and re-annotated segments good enough to serve as a criterion. Our own attempts (three interventions, four negative results, Section 7.3) suggest this is harder than it looks.

An instrument, not a detector — and not yet an AP theory. We do not propose a detector, and the theorems explain why a detector-shaped contribution cannot settle the question this paper asks: a scorer that is constant inside every video keeps 99.7% of the headline number while localizing at chance, so a comparison of two detectors on that number carries no information about the capability the field claims to measure. What we propose instead is an instrument: an estimand, three estimators with their intervals and sensitivity analysis, a partial-identification bound, and a reporting protocol — each a method in the ordinary sense, a procedure that turns labelled scores into a claim that can be checked, and each released as reference code. The four negative results of Section 7.3 are then not a gap in the paper but its evidence: they are what happens when method work is built on the unmeasured quantity. The exchange-rate theorem also does not apply to AP, XD-Violence's headline statistic; empirically AP and AUC almost always agree in *ordering* on every checkpoint we have (Section 4.5), so the qualitative conclusions carry over, but an AP version of Theorem 1 would need a price-weighted decomposition that we leave open.

7.7. Deployment, ethics, and what a diagnostic can and cannot license

Video anomaly detection is a surveillance technology, and the benchmarks we audit are built from a corpus of real surveillance footage. Three things follow from that, none of them comfortable.

The cost of a false alarm is not in the metric. Neither θ, nor B, nor $B|z$ contains a base rate. At the prevalence of anomalous footage in a deployed queue — which is far below the roughly 50:50 balance of a benchmark test split — a detector with high θ can still be wrong on most of the segments it

flags, and no column this paper proposes changes that. A system that selects which video a human reviews should be evaluated with the operating point and the review budget named, not with an AUC.

The label defines the anomaly. "Anomalous" in UCF-Crime and XD-Violence means the categories those annotators chose, labelled at segment granularity by people watching footage from particular places. This paper's own negative result is a statement about that boundary: part of what a trained detector flags inside an anomalous video lies outside the annotated segment, and we cannot tell from the labels whether it is error or genuine anomaly. A claim that a scorer "localizes anomalies well" is therefore always a claim about agreement with one annotation process, and the population it was drawn from is not the population a deployed system would face.

An audit metric is a target once it is adopted. B is a pair-ranking statistic, so it is directly optimisable: a pairwise ranking loss with pseudo-labels — the strategy the companion study tried and that failed under weak labels (Section 7.3) — would improve B without improving detection, and the protocol of Section 5.9 would then be measuring the optimiser rather than the detector. This is the Goodhart problem [37], and it applies to our proposal as much as to AUC. The partial defence is structural: B is an offline diagnostic that needs unit-level ground truth, so it belongs to the benchmark side and cannot be monitored in deployment; and it should be reported next to an *extent* criterion (a TLoC-style detection measure, or average IoU against the annotated segments) so that no single column can be improved alone. We do not claim that defence is sufficient, only that naming it is the minimum a new metric owes.

What a diagnostic of this kind can license is narrow, and we state it in the same spirit as the rest of the paper: it can tell a reader how much of a reported gain could possibly be about localization, and it can tell an author that a claimed gain cannot be one. It cannot certify a detector, and it should not be cited as if it did.

8. Conclusion

Frame-level micro-AUC, the standard headline number of weakly-supervised video anomaly detection, is an exact weighted average of three pairwise contrasts, and the weights are properties of the test split rather than of the method. On UCF-Crime 0.39% of the pairs behind the number are within-video — the only ones that concern localization — and on

XD-Violence 0.07%. The exchange rate follows mechanically: one reported point buys at most 0.0039 points of within-video AUC on UCF-Crime and 0.0007 on XD-Violence. Because the localization term is bounded, the entire attainable range of localization ability moves the reported number by 0.0019, well inside the seed-to-seed noise of 0.0049; no experiment reported at the field's current precision can distinguish a detector that localizes from one that cannot, and a scorer that is constant inside each video — which localizes nothing by construction — retains 93.9% of the state of the art's score.

The weak label does not identify the within-video ordering, which is why post-hoc corrections, including ours, fail; the motion nuisance that dominates the discussion behaves as a mediator rather than a confounder — adjusting for it removes signal — though we state plainly that our design separates the two readings only partially. We therefore propose to *measure* the localization component directly: an unadjusted within-video contrast, a motion-stratified version as a mediator diagnostic, video-clustered intervals, and an E-value, reported alongside the number the field already computes. The cost is one evaluation pass; the benefit is that a paper can say what it actually demonstrated.

Appendix A. Computing the census and the estimators

Everything in Section 5 is arithmetic on the labelled scores of one split. We state the recipe explicitly so that the census and the localization columns can be reproduced from a score file.

Census. For an anomalous video v let a_v and b_v denote its numbers of anomalous and of normal units, M the total number of anomalous units of the split, K the total number of units in normal videos, and B_{tot} the total number of normal units inside anomalous videos. Then

$$n_A = MK, \qquad n_B = \sum_{v \in \mathcal{V}_{\mathrm{an}}} a_v b_v, \qquad n_C = \sum_{v \in \mathcal{V}_{\mathrm{an}}} a_v \, (B_{\mathrm{tot}} - b_v), \tag{A.1}$$

and $\omega_Z = n_Z/(n_A + n_B + n_C)$ for $Z \in \{A, B, C\}$, where $\mathcal{V}_{\mathrm{an}}$ is the set of anomalous videos of the split.

Estimators. With U_v the Mann–Whitney count of video v (one sort of its scores, average ranks for ties),

$$\hat{B}_v = \frac{U_v}{a_v b_v}, \qquad \hat{B} = \frac{\sum_v a_v b_v \, \hat{B}_v}{\sum_v a_v b_v}, \tag{A.2}$$

the pair-weighted version reported throughout. The stratified variant $\hat{B}|z$ replaces U_v by the sum of the per-bin Mann–Whitney counts inside video v, with the bins cut at the within-video quintiles of z. The interval is a bootstrap over videos: resample the V videos with replacement, recompute $\hat{B}$ and $\hat{B}|z$ on the resampled set, and take the 2.5–97.5% quantiles; the unit of replication is the video, not the pair. The E-value is computed from $\hat{B}$ as in Section 5.8, and the partial-identification endpoints as in Section 5.3.

Two implementation notes that matter in practice. First, ties must be broken the way the reported metric breaks them, i.e. by average ranks; ranked scores make the whole estimate one sort per video, so a 290-video split takes about a second. Second, the motion bins must be cut *inside each video*: bins computed on the pooled split would compare a high-motion video's control units with a low-motion video's anomalous units, which is exactly the confounded comparison that the stratification is meant to remove.

Appendix B. Artefacts

We release the reference implementation of the report of Section 5.9, together with the per-frame scores of the sixteen checkpoints used here — the score vector, the per-video lengths, the frame-level ground truth, and the motion statistic — which is sufficient to reproduce every number in Section 6. The archived per-run metric files used for the verification of Section 6.1 are included as well. Evaluation of one checkpoint takes about 16 s on a single RTX A5000 (24 GB) (the released VadCLIP weights take about 55 s, most of it feature extraction); the controlled probes of Section 6.5 train in 30–70 s per arm on 4,000 cached clips.

Appendix C. A residualised sensitivity check, and per-checkpoint bounds

Appendix C.1. A residualised variant, as a sensitivity check only

Fitting the score on z by ridge regression over the control units (those scored below the video median) of one half of the videos and subtracting the fitted value on the other half, with fold assignment by video parity, gives B^{res}. We report it because it is the natural "de-biased" estimator and because it *fails*: on the base checkpoint it lowers B from 0.7558 to 0.7392 and $B|z$ from 0.7166 to 0.7026 (Section 6.5). Under the mediator diagnosis that is the expected outcome, and it is the reason we do not sell it as a method.

Table C.8: Partial identification of the within-video contrast for every checkpoint. $\pi_{\text{mot}} = (B - B|z)/(1 - B|z)$ is the motion-mediated share of the headroom above the stratified estimator (a mediation proportion, not a contamination share; Section 5.3); $\delta^* = 2B - 1$ is the contamination share at which the localization claim dies; $\tau_{\text{LB}}(0.2)$ is the lower endpoint of the identified interval under 20% contamination.

checkpoint	B	$B\|z$	π_{mot}	δ^*	$\tau_{\text{LB}}(0.2)$
UCF base s234	0.7270	0.6729	0.165	0.454	0.6587
UCF motion6 s789	0.7576	0.7230	0.125	0.515	0.6970
UCF base s789	0.7558	0.7166	0.138	0.512	0.6948
UCF cons0 s234	0.7235	0.6744	0.151	0.447	0.6544
UCF attr s234	0.7080	0.6545	0.155	0.416	0.6351
UCF core14txt s234	0.7169	0.6728	0.135	0.434	0.6462
UCF motion6 s555	0.7180	0.6604	0.170	0.436	0.6476
UCF base s555	0.7236	0.6688	0.165	0.447	0.6546
UCF core8g1 s234	0.7124	0.6621	0.149	0.425	0.6406
UCF res s234	0.7467	0.7093	0.128	0.493	0.6833
UCF motion6-shuf s234	0.7171	0.6786	0.120	0.434	0.6463
XD base s234	0.8466	0.8223	0.137	0.693	0.8083
XD base s555	0.8419	0.8160	0.141	0.684	0.8024
XD base s789	0.8281	0.8000	0.140	0.656	0.7851
XD motion3 s789	0.8046	0.7774	0.122	0.609	0.7557

Appendix C.2. Per-checkpoint partial identification

Appendix D. The released VadCLIP weights as an external anchor

The twelfth row of Table 6 is the checkpoint released with VadCLIP, scored by this paper's suite rather than by its own authors: weights load without a missing or unexpected key. Two AUC conventions appear here, and we keep them separate. The repository's own evaluation path scores frames with the per-snippet score repeated 16×; under that convention the released weights reproduce the published frame-level AUC (0.8808 against the 0.8802 the paper reports). This paper's census and its eleven-arm comparison are computed on the snippet grid — one score and one label per sixteen frames — where the same weights give $\theta = 0.8560$, and it is this snippet-grid value that enters Table 6 and the band comparison; the 0.025 difference between the two conventions is a unit-of-analysis difference, not a reproduction failure. On the snippet grid θ is 0.0056 below the weakest of our arms, which is within the 0.0049 between-seed standard deviation of θ, and its $(\theta, B, B|z) = (0.8560, 0.7157, 0.6713)$ land inside the band spanned by our

eleven arms ($B \in [0.708, 0.758]$, $B|z \in [0.655, 0.723]$). The scores are the model's own head, not our refinement, and the pair census comes out at the same $\omega = 0.6317$, so the comparison is between two independent score pipelines rather than two settings of one. The band therefore brackets the published model rather than being a self-selected set that happens to exclude it; its mAP@0.5 under that detection protocol (2.95) sits inside the family's span $[2.44, 5.82]$, while its AP@0.5 (1.18) falls below the family's minimum (1.85). That is the sense in which the localization columns are comparable across implementations — not that it predicts them.

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